Cyclotomy
The Story of How Gauss Narrowly Missed Becoming a Philologist

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Abstract

It was known to ancient Greek geometers that one can construct an equilateral triangle, a square, a regular pentagon, and a regular hexagon using only a ruler and compass (*convince yourself*!). But in **1796**, a teenage prodigy named **Carl Friedrich Gauss** stunned the mathematical world by showing that a **regular 17-gon** could also be constructed with just a compass and straightedge — a feat untouched since the time of the Greeks. This moment, immortalized by Gauss with a single triumphant entry in his diary, marked the beginning of a deep and beautiful theory: **cyclotomy**, the division of a circle into equal parts.

Behind this elegant geometry lies a story of unexpected connections — between **algebra and number theory**, between **roots of unity and modular arithmetic**, and, perhaps most surprisingly, between **mathematics and philology**. In this article, we'll explore — through narration and perhaps a little mathematics — how Gauss's pursuit of cyclotomic equations led to the foundations of modern algebra, and how his curiosity about **language** and **structure** nearly drew him into the world of philology instead.

Join with me for an incredible journey spanning **Greece through Aachen**!

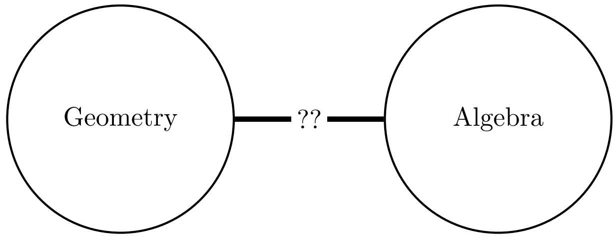
1 Introduction

Cyclotomy- literally means circle cutting was a puzzle begun by the Greek geometers, more than 2000 years ago. In this pastime, they used two implements - a ruler (straightedge) to draw straight lines and a compass to draw circles. (Probably they had lot of spare time since there were no mobiles or televisions back then!!!). The problem of cyclotomy was to divide the circumference of a circle into n equal parts using only these two implements.

As these n points on the circle are also the corners of a regular n-gon, the problem of cyclotomy is equivalent to the problem of constructing the regular n-gon using only a ruler and a compass. It's no surprise that Euclid's school was interested in this problem and constructed the equilateral triangle, the square, the regular pentagon and the regular hexagon. For more than 2000 years mathematicians had been unanimous in their view that for no prime p bigger than 5 can the p-gon be constructed by ruler and compasses. The teenager Carl Friedrich Gauss proved a month before he was 19!! that the regular 17-gon is constructible. He did not stop there but went ahead to completely characterise all those n for which the regular n-gon is constructible! This achievement of Gauss is one of the most surprising discoveries in mathematics. This feat was responsible for Gauss dedicating his life to the study of mathematics instead of philology, in which too he was equally proficient.

He was so proud of this discovery that he requested that the regular 17-gon be engraved on his tombstone! Unfortunately that wish was not carried out. Gauss's proof was the first instance when a mathematical problem from one domain was rephrased in another domain and solved successfully. Another instance of this we saw in the proof of the "Fundamental Theorem of Algebra" using complex analysis. In this instance, the geometrical problem of cyclotomy was recast in algebraic terms and solved!

So there would be 3 parts of this article-



Understanding the original version of the problem i.e geometric which we did through the story, then how the algebraic reformulation is made which is the next part and finally understanding the "question marked" bridge that links this two notions.

Now lets see in details what cyclotomy is all about algebraically. For that we need to change our viewpoint and introduce some algebraic notion. To do thiswe turn to the methods of analytic geometry but before that we need to see some standard construction from elementary geometry that we state-

- Drop a perpendicular on a given line l from a point P outside it.Draw a circle centred at P cutting l at A and B. Draw circles centred at A and B having radii AP and BP, respectively. The latter circles intersect at P and Q and PQ is perpendicular to l. See the figure (Figure 1) below

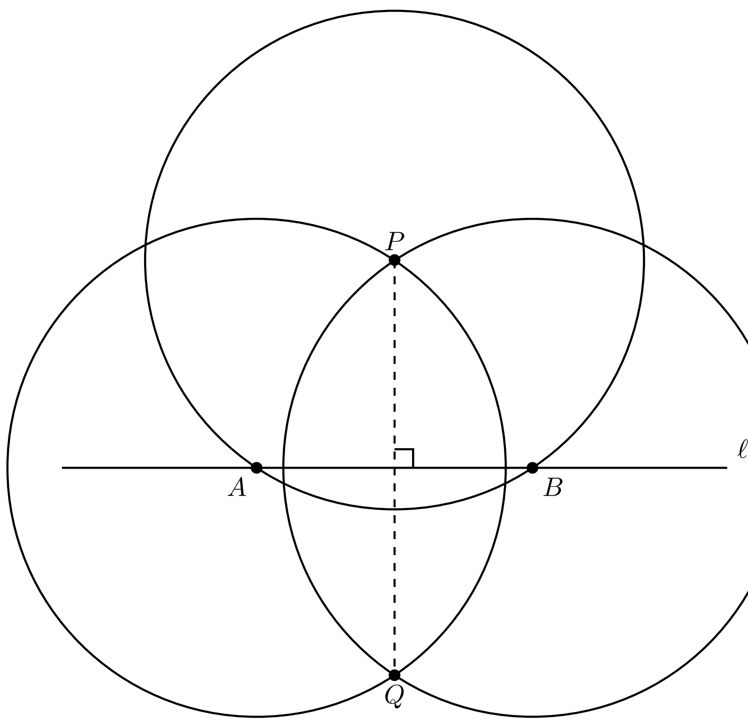


Figure 1: Drop a perpendicular from P to ℓ

- Draw a perpendicular to a given line l through a point P on it. Draw any circle centred at P intersecting l at A and B. Then, the circles centred at A and B with the common radius AB intersect at two points C and D. Then, CD passes through P and is perpendicular to l. See the figure (Figure 2) below
- Draw a line parallel to a given line through a point outside.This follows by doing the above two constructions in succession.
- Bisect a given segment AB. This is obvious from the second point.

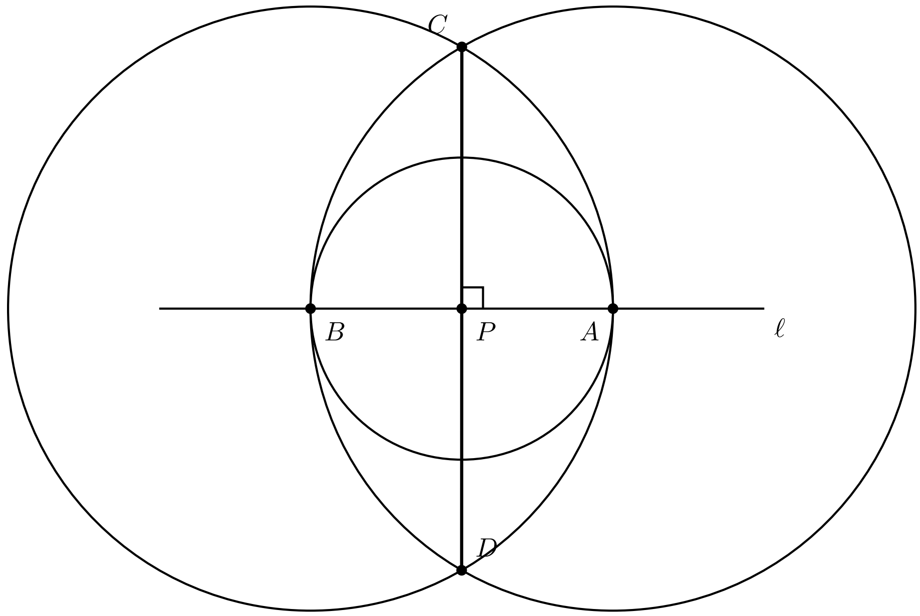


Figure 2: Construct a perpendicular to ℓ through P

2 Some Algebra

Now we go back to our discussion to set up a coordinate axis. Given our original two points, we set up a coordinate system by defining X-axis as the line through two points. Setting one point to be origin, and the other as (1,0). We can draw line perpendicular to X-axis through the origin and we can call that to be Y-axis. (See second point above).

Thus we see marking the point (a, b) on the plane is, by these observations, equivalent to the construction of the lengths $|a|$ and $|b|$. We say $a \in \mathbb{R}$, constructible if we can construct 2 points at a distance $|a|$ apart. Equivalently a is constructible if either $(a, 0)$ or $(0, a)$ is constructible. Further we can view the construction of (a, b) as construction of the complex number $a+ib$. We see that if a,b are constructible numbers then so is $a+b$, $a-b$, ab, a/b. We give the constructions for $a+b$ (Figure 3) and ab (Figure 4), since the constructions for $a-b$ and a/b are identical. See the figure below.

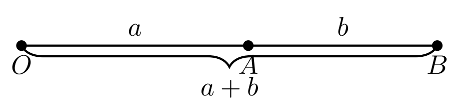


Figure 3: Construction of line segment of $a + b$.

This makes the set of constructible numbers a subfield of $\mathbb{R}$. We also have if a is constructible so is $\sqrt{a}$. See the figure (Figure 5) below.

It's not hard to see by the four points above (a, b) is constructible if and only if $(a, 0)$ and $(0, b)$ constructible or in other words a and b are constructible. In other words in order to construct a number c, we must draw a finite number of circles and lines in such a way so that $|c|$ is the distance of intersection or equivalently we must draw circles and lines so that $(c, 0)$ is a point of intersection.

Remark 1. Let $\mathbb{K}$ be the field generated over $\mathbb{Q}$ by all numbers obtained in some such construction, we obtain a subfield of the field of constructible numbers. To give a criterion for when a number is constructible we need to relate constructiblity to the properties of the extension $\mathbb{K}/\mathbb{Q}$.

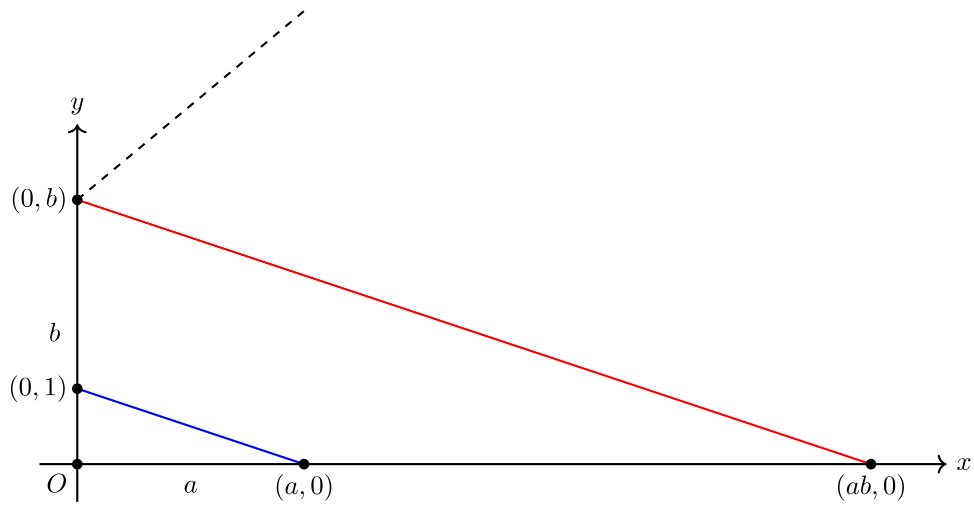


Figure 4: Construction of ab. Join line between $(0, 1)$ and $(a, 0)$ and then draw a line parallel to the joined line through $(0, b)$ and let it intersect the X-axis at a point. Then by similarity of triangles one can show the coordinate is $(ab, 0)$

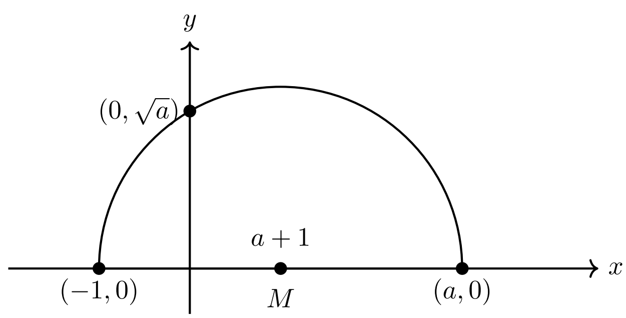


Figure 5: Construction of $\sqrt{a}$. It's an elementary calculation using Pythagoras theorem that the coordinate of intersection of the semicircle and the Y-axis is indeed $(0, \sqrt{a})$

Remark 2. Since by the coordinate geometry, a line and a circle have equations of the form $ax + by - c = 0$, $(x - s)^2 + (y - t)^2 = r^2$, their intersection (if any) are the common roots. Eliminating one of x, y we get quadratic for other. Thus use of ruler and compass amounts in algebraic terms to solving a chain of quadratic equations.

Before we move to the technicalities we see some concrete examples of our last remark!! By all these discussions we have reduced the problem of cyclotomy to

1. finding roots of the equation $z^n - 1 = 0$ or,
2. Construct the angle $2\pi/n$ or equivalently $\cos(2\pi/n)$ or $\sin(2\pi/n)$ are constructible numbers. See the figure (Figure 6) below.

For n=2, one needs to draw a diameter and this is evidently achieved by ruler.

For n=3, the equation $z^3 - 1 = 0$ reduces the equations $z - 1$ or $z^2 + Z + 1 = 0$. The roots of the latter are $\frac{-1+\sqrt{3}i}{2}$ and $\frac{-1-\sqrt{3}i}{2}$. So we have to bisect the line segment [-1,0] and the points of intersection of this bisector with the unit circle are the points we want to mark off the circle. See figure (Figure 7)

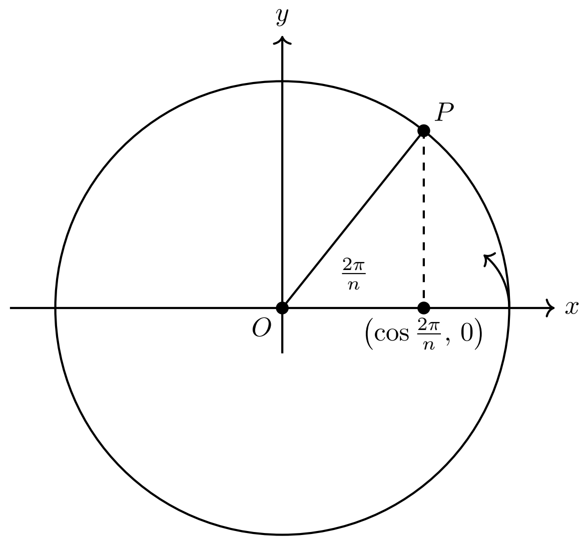


Figure 6: This shows why marking off an angle of $2\pi/n$ is equivalent saying that $\cos(2\pi/n)$ is a constructible number.

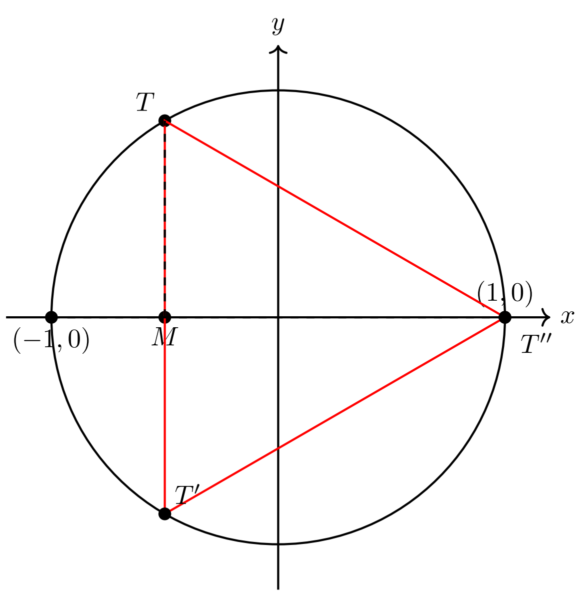


Figure 7: Constructing an equilateral triangle in the unit circle.

For n=4, we only need bisection of the X-axis is involved. This demonstrates that if a regular n- gon is constructible so is $2^r n$-gon for any $r \in \mathbb{N}$. We just keep bisecting each sides. So in particular 2^r-gons are constructible.

To construct the regular pentagon, one has to construct the roots of $z^5 - 1 = 0$. They are the 5th roots of unity ζ^k, $1 \leq k \leq 5$ and $\zeta = e^{2\pi i/5}$. Now $\zeta + \zeta^2 + \zeta^3 + \zeta^4 + \zeta^5 = 0$, using $\zeta^5 = 1$ we get $(\zeta^2 + \zeta^3)(\zeta + \zeta^4) = \zeta + \zeta^2 + \zeta^3 + \zeta^4 = -1$. This gives us that $\zeta^2 + \zeta^3$ and $\zeta + \zeta^4$ are the two roots of the quadratic polynomial $x^2 + x - 1 = 0$. Set $\zeta + \zeta^4$ (being positive) equal to $\frac{-1+\sqrt{5}}{2}$ and multiplying by ζ and using $\zeta^5 = 1$ we get a quadratic equation for ζ!! This is the algebraic reasoning behind it's construction. If this is not convincing the following lemma might help us.

Lemma 1. *If a and b are constructible real numbers, then the roots of the polynomial $x^2 - ax + b = 0$ are also constructible*

Proof. We take the points (a, b) (such a point exists since a and b are constructible) and $(0, 1)$ and draw a circle with these two points as diameter. Now see it's intersection with the X-axis and these two points are our required roots of the polynomial. See the figure below

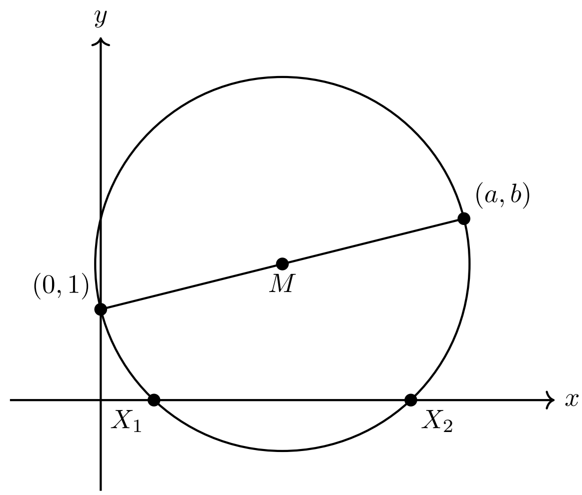


Figure 8: Circle with diameter $(0, 1)$–(a, b) and intersections with the X-axis are our required points.

□

In our case for n=5 we have that ζ satisfies a quadratic equation namely $x^2 - (\frac{-1+\sqrt{5}}{2})x + 1 = 0$ and since the set of constructible numbers forms a subfield enough to check that $\sqrt{5}$ is constructible which is obvious. Thus by our previous lemma we see that we can construct ζ!!

Now for the famous construction of Gauss for 17-gon, we denote $\zeta = e^{2\pi i/17}$. Then $\zeta^{17} = 1$ gives $\zeta^{16} + \zeta^{15} + ... + \zeta + 1 = 0$. Let us consider three real numbers α_1, α_2, α_3 defined as follows
$\alpha_1 := \zeta^3 + \zeta^{-3} + \zeta^5 + \zeta^{-5} + \zeta^6 + \zeta^{-6} + \zeta^7 + \zeta^{-7}$
$\alpha_2 := \zeta^3 + \zeta^{-3} + \zeta^5 + \zeta^{-5}$
$\alpha_3 := \zeta + \zeta^{-1}$
Now we look at the sequence of the four quadratic equations

$$x^2 - \alpha_3 x + 1 = 0 \tag{1}$$

$$x^2 - (\frac{\alpha_2^3 - 6\alpha_2 - 2\alpha_1 + 1}{2})x + \alpha_2 = 0 \tag{2}$$

$$x^2 - \alpha_1 x - 1 = 0 \tag{3}$$

$$x^2 - x - 4 = 0 \tag{4}$$

A bit of calculation !! shows that $\zeta, \alpha_3, \alpha_2, \alpha_1$ are the roots of these quadratic equations in the order(1 to 4). Thus by our previous lemma [Lemma 1] we can see that ζ is recursively constructible!!

Now the question remains as to what made this work and which other n-gons are constructible or as would be evident from the calulation above ζ can be expressed as nested chain of square roots. For example the last calculation shows that

$$\cos(2\pi/17) = -\frac{1}{16} + \frac{1}{16}\sqrt{17} + \frac{1}{16}\sqrt{34 - 2\sqrt{17}} + \frac{1}{8}\sqrt{17 + 3\sqrt{17} - \sqrt{34 - 2\sqrt{17}} - 2\sqrt{34 + 2\sqrt{17}}}$$

From our discussion so far it seems that for a prime p if $p = 2^k + 1$ then we have some hope of p-gon being constructible. Indeed that's what we would prove in the next section, (we need to formalize this naive idea). It's easy to see that for p to be prime k has to be power of 2. These are called Fermat's prime after Pierre de Fermat as he was the one who investigated the sequence. Fermat thought that the numbers $2^{2^n} + 1$ are primes for all n. However, the only primes of this form found until now are 3, 5, 17, 257 and 65537(!) The number $2^{2^5} + 1$ was shown by Euler to have 641 as a proper factor.

Till now everything was our thought. Now we need a language to express those, just as when we write a letter to someone, we have some thoughts and then we need a language to write it down. So if someone don't get anything it's fine, it's just formalizing whatever we have done so far. we need the language of Galois Theory which would act as a bridge between these two realms formalizing our naive idea. But before that we need to talk about cyclotomic fields and cyclotomic polynomials as that would be the building blocks of our bridge.

But before we start to lighten the environment we digress a bit to another relevant story. We all heard about the name Abel. He earned fame by proving that the general equation of degree atleast five is not solvable by a 'formula' involving only square roots, cube roots and higher roots. One of his lesser-known achievements involves a problem analogous to cyclotomy viz., the division of the lemniscate. The name lemniscate literally means a ribbon and comes from its shape; this curve - also called the elastic curve - was discovered by Bernoulli. See figure below

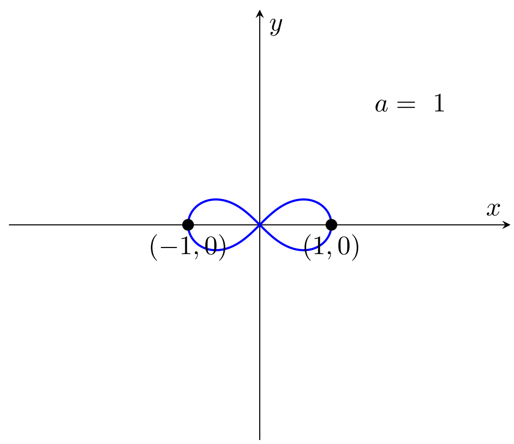


Figure 9: Bernoulli lemniscate given by $r^2 = a^2 \cos(2\theta)$.

3 Building the bridge

Let $\zeta_n = e^{2\pi i/n}$ be a primitive nth root of unity. Then we know $\mathbb{Q}(\zeta_n)$ is called cyclotomic field over $\mathbb{Q}$ (this is obtained by adjoining a primitive nth root of unity to $\mathbb{Q}$). The elements $\{1, \zeta, ..., \zeta^{\varphi(n)-1}\}$ forms a $\mathbb{Q}$-basis for $\mathbb{Q}(\zeta_n)$. Thus each element of $\mathbb{Q}(\zeta_n)$ is some $\mathbb{Q}$-linear combination of $\{1, \zeta, ..., \zeta^{\varphi(n)-1}\}$.
The nth cyclotomic polynomial is defined as

$$\Phi_n(x) := \prod_{j=1}^{r} (x - \zeta_j) \;\; where \;\; \zeta_j = e^{2i\pi j/n} \;\;,\;\; gcd(j, n) = 1$$

Remark 3. An interesting feature of the cyclotomic polynomials is the following. The coefficients seem to be among 0, 1 and -1 for instance, for a prime p, $\Phi_p(x) = 1 + x + x^2 + ... + x^{p-1}$ and one might wonder whether this is true for any $\Phi_n(x)$. It turns out that $\Phi_{105}(x)$ has one coefficient equal to 2! Using, some nontrivial results on how prime numbers are distributed, one can show that every integer occurs among the coefficients of the cyclotomic polynomials!

We now use cyclotomic polynomials to give a proof for a very special case of Dirichlet's theorem on primes in arithmetic progression.

Proposition 1. *Let $d \in \mathbb{N}$ and $d \geq 2$. Then there are infinitely many primes of the form $nd + 1$.*

Proof. We need the following lemma for finishing the problem.

Lemma 2. *Let $d \in \mathbb{N}$ be fixed natural number and let $b \in \mathbb{N}$ is arbitrary natural number. Let p be a prime number such that $p \mid \Phi_d(b)$ then either $p \mid d$ or $p \equiv 1 \pmod{d}$*

Proof. See [7] for the proof ☐

If we can show that $S = \{p | p \mid \Phi_d(b)\}_{b\in\mathbb{N}}$ is infinite then by previous lemma we get that there are infinitely many primes of the form $nd+1$. Say $|S| < \infty$, say $S = \{p_1, p_2, ..., p_r\}$. Then by (equation 5) we get that for every $b \in \mathbb{N}$ $\exists p_i \in S$ such that $p_i \mid (b^n - 1)$. But taking $b = \prod p_i$ gives us a contradiction. This proves that $|S| = \infty$ and consequently this shows that there are infinitely many primes of the form $nd + 1$. ☐

Euclid's classical proof of the infinitude of prime numbers is the special case of the above proof where we can use d = 2.
There is no reason to believe that $\Phi_n(x) \in \mathbb{Z}[X]$ but using elementary number theory one can show that

$$x^n - 1 = \prod_{d|n} \Phi_d(x) \tag{5}$$

and then use induction to prove that $\Phi_n(x) \in \mathbb{Z}[X]$. One can easily see that $deg(\Phi_n(x)) = \varphi(n)$ where $\varphi(n)$ is the Euler-totient function. One can also show that $\Phi_n(x)$ is actually irreducible in $\mathbb{Q}[X]$ that is to say that $\Phi_n(x)$ can't be written as a product f(x) and g(x) with $deg(f(x))$ and $deg(g(x))$ less than $\varphi(n)$. Thus $\Phi_n(x)$ serves as so called the minimal polynomial for ζ_n. The experts, can show $Gal(\mathbb{Q}(\zeta_n)/\mathbb{Q}) = (\mathbb{Z}/n\mathbb{Z})^*$.

Now the next lemma formalizes our previous statement "Thus use of ruler and compass amounts in algebraic terms to solving a chain of quadratic equations" and gives us a connection between constructibility and field extensions.

Lemma 3. *Let $\mathbb{K}$ be a subfield of $\mathbb{R}$*

1. *The intersection of two lines is either empty or is a point in the plane of $\mathbb{K}$.*
2. *The intersection of a line and a circle in $\mathbb{K}$ is either empty or consists of one or two points in the plane of $\mathbb{K}(\sqrt{\theta})$ for some $\theta \in \mathbb{K}$ with $\theta \geq 0$.*
3. *The intersection of two circles in $\mathbb{K}$ is either empty or consists of one or two points in the plane of $\mathbb{K}(\sqrt{\theta})$ for some $\theta \in \mathbb{K}$ with $\theta \geq 0$.*

Proof. The first point is an easy calculation. For the remaining two it's sufficient to prove the second statement, since if $x^2+y^2+ax+by+c = 0$ and $x^2+y^2+a'x+b'y+c' = 0$ are the equations of circles C and C', respectively, then their intersection is the intersection of C with the line $(a-a')x+(b-b')y+(c-c') = 0$. So, to prove the statement 2, suppose our line L in $\mathbb{K}$ has the equation $dx + ey + f = 0$. We may assume $d \neq 0$, since if $d = 0$, we have $e \neq 0$. By dividing with d, we may assume that $d = 1$. Plugging $-x = ey + f$ in the equation of C we get,

$$(e^2 + 1)y^2 + (2ef - ae + b)y + (f^2 - af + c) = 0$$

We identify this equation with $\alpha y^2 + \beta y + \gamma = 0$. If $\alpha = 0$ then $y \in \mathbb{K}$, else by completing the square we get that either $L \cap C = \phi$ or $y \in \mathbb{K}(\sqrt{\beta^2 - 4\alpha\gamma})$ with $\beta^2 - 4\alpha\gamma \geq 0$. ☐

From this lemma we can see that we turn the definition of constructibility into a property of field extension of $\mathbb{Q}$, and in doing so we obtain a criterion for when a number is constructible. In words the lemma says that while constructing a constructible number over a field $\mathbb{K}$, we either remain in the same field $\mathbb{K}$ or go to some real quadratic extension of our field $\mathbb{K}$.
Now we state possibly the most important theorem of the whole article on which lies the solution to the problem of constructibility.

Theorem 1. *A real number c is constructible if and only if there is a tower of fields* $\mathbb{Q} = \mathbb{K}_0 \subseteq \mathbb{K}_1 \subseteq ... \subseteq \mathbb{K}_r$ *such that* $c \in \mathbb{K}_r$ *and* $[\mathbb{K}_{i+1} : \mathbb{K}_i] \leq 2$ *for each i. Therefore if c is a constructible number, then c is algebraic over* $\mathbb{Q}$ *and* $[\mathbb{Q}(c) : \mathbb{Q}] = 2^r$ *for some* $r \in \mathbb{N}$.

Proof. If c is constructible, then the point $(c, 0)$ can be obtained from a finite sequence of constructions starting from the plane of $\mathbb{Q}$. We then obtain a finite sequence of points, each an intersection of constructible lines and circles, ending at $(c, 0)$. By [Lemma 2], the first point either lies in $\mathbb{Q}$ or in $\mathbb{Q}(\sqrt{\theta})$ for some θ. This extension has degree either 1 or 2. Each time we construct a new point, we obtain a field extension whose degree over the previous field is either 1 or 2 by the lemma. Thus, we obtain a sequence of fields

$$\mathbb{Q} = \mathbb{K}_0 \subseteq \mathbb{K}_1 \subseteq \mathbb{K}_2 \subseteq ... \subseteq \mathbb{K}_r$$

with $[\mathbb{K}_{i+1} : \mathbb{K}_i] \leq 2$ and $c \in \mathbb{K}_r$. Therefore, $[\mathbb{K}_r : \mathbb{Q}] = 2^n$ for some n. However $[\mathbb{Q}(c) : \mathbb{Q}]$ divides $[\mathbb{K}_r : \mathbb{Q}]$ (since $\mathbb{Q}(c) \subseteq \mathbb{K}_r$), so $[\mathbb{Q}(c) : \mathbb{Q}]$ is also a power of 2.
For the converse , suppose that we have a tower $\mathbb{Q} = \mathbb{K}_0 \subseteq \mathbb{K}_1 \subseteq \mathbb{K}_2 \subseteq ... \subseteq \mathbb{K}_r$ with $c \in \mathbb{K}_r$ and $[\mathbb{K}_{i+1} : \mathbb{K}_i] \leq 2$ for each i. We show that c is constructible by induction on r. If $r = 0$ then $c \in \mathbb{Q}$ and hence constructible. Assume that $r > 0$ and that the elements of $\mathbb{K}_{r-1}$ are constructible. Since $[\mathbb{K}_r : \mathbb{K}_{r-1}] \leq 2$, the quadratic formula shows that we may write $\mathbb{K}_r = \mathbb{K}_{r-1}(\sqrt{\alpha})$ for some $\alpha \in \mathbb{K}_{r-1}$. Since α is constructible we have that $\sqrt{\alpha}$ is also constructible. Therefore, $\mathbb{K}_r = \mathbb{K}_{r-1}(\sqrt{\alpha})$ lies in the field of constructible numbers and hence, c is constructible.

□

Now we are in a position to tell for which $n \in \mathbb{N}$ are regular n-gons constructible.. which we state in form of the next and our last theorem.

Theorem 2. *A regular n-gon is constructible if and only if* $\varphi(n)$ *is a power of 2*

Proof. We have already seen that a regular n-gon is constructible if and only if the central angles $2\pi/n$ are constructible and we have seen this occurs if and only if $\cos(2\pi/n)$ is a constructible number. Let $\zeta = e^{2i\pi/n} = \cos(2\pi/n) + i\sin(2\pi/n)$, a primitive nth root of unity. Then $\cos(2\pi/n) = \frac{1}{2}(\zeta + \zeta^{-1})$, since $\zeta^{-1} = \cos(2\pi/n) - i\sin(2\pi/n)$. Thus $\cos(2\pi/n) \in \mathbb{Q}(\zeta)$. However $\cos(2\pi/n) \in \mathbb{R}$ and $\zeta \notin \mathbb{R}$, so $\mathbb{Q}(\zeta) \neq \mathbb{Q}(\cos(2\pi/n)$. But ζ is a root of the quadratic polynomial $x^2 + 2\cos(2\pi/n)x + 1$, so we get $[\mathbb{Q}(\zeta) : \mathbb{Q}(\cos(2\pi/n)] = 2$. Therefore if constructible then $[\mathbb{Q}(\cos(2\pi/n) : \mathbb{Q}]$ is a power of 2.
Hence $\varphi(n) = [\mathbb{Q}(\zeta) : \mathbb{Q}]$ is also a power of 2.
Conversely, suppose that $\varphi(n)$ is a power of 2. The field $\mathbb{Q}(\zeta)$ is a Galois extension of $\mathbb{Q}$ with an abelian Galois group (as we have remarked earlier it's Galois group is $(\mathbb{Z}/n\mathbb{Z})^*$. If $H = Gal(\mathbb{Q}(\zeta) : \mathbb{Q}(\cos(2\pi/n)))$, we can use induction and the following fact from group theory that every finite 2-group has a subgroup of index 2 to conclude that there is a chain of subgroups

$$H_0 \subseteq H_1 \subseteq ... \subseteq H_r = H$$

with $|H_{i+1} : H_i| = 2$. If $L_i = \Gamma(H_i)$, i.e the fixed field of H_i, then $[L_i : L_{i+1}] = 2$; and thus $L_i = L_{i+1}(\sqrt{u_i})$ for some u_i. Since $L_i \subseteq \mathbb{Q}(\cos(2\pi/n) \subseteq \mathbb{R}$, each of the $u_i \geq 0$ otherwise it would not have been a subfield of $\mathbb{R}$. Since the square root of a constructible number is constructible we see that everything in $\mathbb{Q}(\cos(2\pi/n)$ is constructible by [Theorem 1]. Thus we see that $\cos(2\pi/n)$ is constructible and hence a regular n-gon is constructible. □

This concludes our long journey from "Greece to Aachen" through GT route!!

But before ending this article we would like to see some examples of irreducible polynomials over $\mathbb{Q}[X]$ because it's interesting in itself and a topic to study of it's own. Irreducibility over $\mathbb{Q}$ is particularly interesting as that gives us a finite field extension of $\mathbb{Q}$ which is of importance in the study of number theory in particular and Galois theory in general.

1. (Selmer) The enthusiasts may try to show that the polynomial $x^n - x - 1$ is irreducible in $\mathbb{Q}[X]$ for all $n \in \mathbb{N}$. The experts can also try to show that it's Galois group is actually full! (i.e it's S_n).

2. (Cohn) Also I would like to mention one more cute example which I happened to come across recently. If we take any prime number of our choice and take any base $b \geq 2$ of our choice. We now write the prime in the base b. We collect the coefficients and make a polynomial out of it. The resulting polynomial would be irreducible in $\mathbb{Q}[X]$!!

For example if we take 31 as prime and base as 3. we get $31 = 1.3^3 + 1.3 + 1$ so the corresponding polynomial is $x^3 + x + 1$ which by Gauss lemma is enough to show irreducible in $\mathbb{Z}[X]$, which is easy to show. We may work with any other prime and any other base of our choice.
Thus we have a plethora of irreducible polynomials over $\mathbb{Q}[x]$ at our disposal. Isn't it beautiful!!!!

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